

COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4 · MODULE 6

# Application Protocols: the Web, Email and DNS

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · DNS

Module 6 – Application Protocols

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OUTLINE

# Module 6 — roadmap

**General concepts about network applications**

**One network application: the Web**

**HTTP: the protocol that supports the Web**

**Email**

**DNS: the Domain Name System**

MODULE 6

# Network Applications

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What an application needs from the layers beneath it

1 • Network applications

2 • The Web and HTTP

3 • Email

4 • DNS

# What is a network application?

**Programs that run on different end systems** and communicate with each other over the network.

## **Web browsing**

Google Chrome, Mozilla Firefox,  
Microsoft Edge

## **Email**

Gmail, Outlook

## **Instant messaging**

Slack, WhatsApp, Facebook  
Messenger

## **Streaming services**

YouTube, Netflix

## **Social media**

Instagram, TikTok

## **AI assistants**

Claude, ChatGPT, Gemini

# Application-layer protocols

An **application-layer protocol** is one piece of a network application — but a big piece.

It defines how the application's processes, running on different end systems, **pass messages to each other**: which messages exist, their format, and when and how to respond.

This is what lets programs on different machines work together — e.g. any web browser can fetch a page from any web server, whoever wrote each program.

| Network application | Application-layer protocol |
|---------------------|----------------------------|
| Web application     | HTTP                       |
| Email               | SMTP                       |

# Choosing a transport protocol

- Application-layer protocols are controlled by the **application developer**; transport-layer protocols, however, are built into the **OS**.
- As a developer you can only choose between the transport protocols the OS provides — so pick the one that best fits what your application-layer protocol needs.

| Property               | Example applications                                | Transport protocol                                      |
|------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Reliable data transfer | Email, IM, file transfer, financial apps            | <b>TCP</b> preferred<br>UDP unsuitable                  |
| Low delay              | Internet telephony, conferencing, multiplayer games | <b>UDP</b> preferred<br>TCP retransmissions add latency |

MODULE 6

# The Web and HTTP

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The application-layer protocol behind every page you load

1 · Network applications

2 · **The Web and HTTP**

3 · Email

4 · DNS

# The World Wide Web

## THE WEB

The **World Wide Web** (aka Web) is a **network application**: a global collection of documents and other resources shared over the Internet — like a large file storage system.

Each object on the Web (e.g., an image, an HTML file) is addressable by a **URL**:

```
http://www.someschool.edu/someDept/pic.gif  
└─ host name ─┐ └─ path name ─┘
```

## HTTP

# HTTP: the protocol behind the Web

### HTTP

The **Hypertext Transfer Protocol** is the **application-layer protocol** that supports the Web — it defines how clients request objects by URL and how servers respond.

HTTP follows a **request–response** pattern: the client sends an **HTTP request** for an object, and the server answers with an **HTTP response** carrying it.



HTTP

# How HTTP works

HTTP uses **TCP** as its transport protocol — so before any HTTP message is sent, a TCP connection must be established first.



## STATE LESS

HTTP is **stateless** — the server maintains no information about past client requests. Each request is handled independently.

# HTTP connections

## Non-persistent HTTP

- At most **one object** is sent over a TCP connection — connection is then closed
- HTTP/1.0 uses non-persistent HTTP

## Persistent HTTP

- **Multiple objects** can be sent over a single TCP connection
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP

| Protocol | Published | Connection type   | Status   |
|----------|-----------|-------------------|----------|
| HTTP/1.0 | 1996      | Non-persistent    | Legacy   |
| HTTP/1.1 | 1997      | Persistent        | Standard |
| HTTP/2   | 2015      | Persistent        | Standard |
| HTTP/3   | 2022      | Persistent (QUIC) | Standard |

# Non-persistent HTTP in action

Suppose a client wants to retrieve a file from: `http://www.someSchool.edu/someDepartment/home.index`

1. Client establishes a **TCP connection** with the server
2. Client sends an **HTTP GET** request — path: `/someDepartment/home.index`
3. Server retrieves the HTML file, wraps it in an **HTTP response**, and sends it back
4. Server signals TCP to **close the connection**



# Response time: non-persistent HTTP

## RESPONSE TIME

The time to retrieve a single object using HTTP — from the client's request until the object is received

**Q:** What is the response time for non-persistent HTTP?

**A:** 2 RTTs + file transmission time

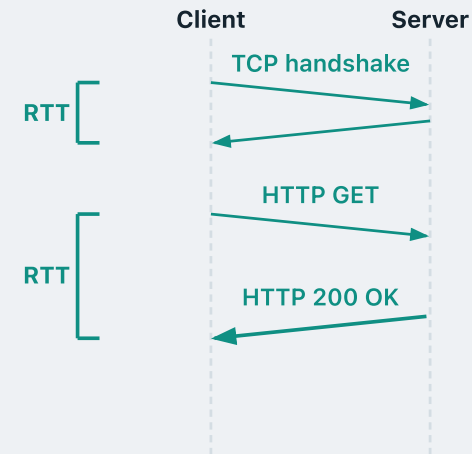
## RTT

Recall: **round-trip time** — the time for a small packet to travel from client to server and back

**First RTT:** the TCP handshake — client sends SYN, server replies SYN-ACK

**Second RTT:** client sends HTTP GET, server sends back the response

With many referenced objects, this cost multiplies — a significant burden on the server



## Persistent HTTP in action

Client and server establish a TCP connection and exchange HTTP messages

Server **keeps the TCP connection open** after sending the first HTTP response

Subsequent HTTP messages between the same client and server use the **same TCP connection** — no new handshake needed

### Response time on one persistent connection:

1 RTT for the initial TCP handshake, then 1 RTT for each HTTP request.

So for  $n$  requests: **total**  $\approx (1 + n)$  RTT (plus transmission time).

Amortised over the  $n$  objects:  $(1 + n)/n = (1/n + 1)$  RTT. As  $n$  becomes large,  $1/n \rightarrow 0$ , so this approaches **~1 RTT per object**.

Can we do even better than 1 RTT per object?

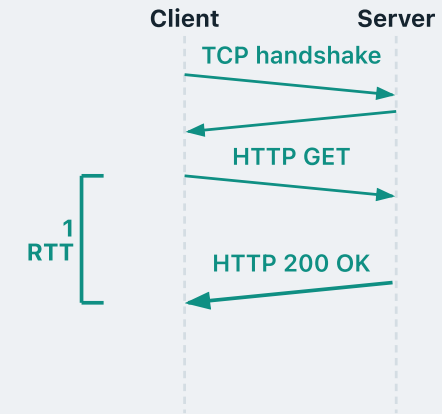


HTTP

## Improving HTTP efficiency further with pipelining

**WITHOUT  
PIPELINING**

Client issues a new request **only after** receiving the previous response —  
**1 RTT per file**



# Improving HTTP efficiency further with pipelining

## WITHOUT PIPELINING

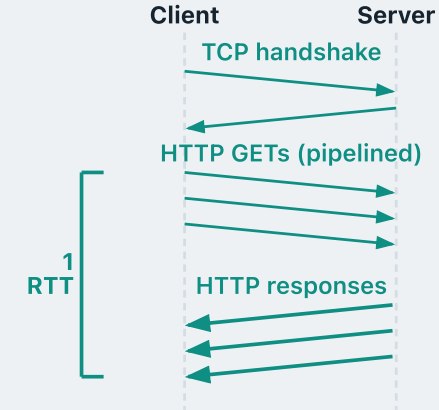
Client issues a new request **only after** receiving the previous response — **1 RTT per file**

## WITH PIPELINING

Client sends requests **as soon as it needs to retrieve an object**, without waiting for responses — e.g. in this example we pipeline 3 HTTP requests all together

**Benefit:** pipelining 3 requests → response time per file  $\approx 1 \text{ RTT} / 3$

- Don't confuse this with TCP-level pipelining.
- **HTTP pipelining** operates at the **application layer**: the client sends several **HTTP requests** without waiting for each response.
- **TCP pipelining** operates at the **transport layer**: the sender keeps several **segments** in flight within its window. TCP cuts each HTTP message into segments, so these may be pieces of one HTTP request, or of several.



# HTTP request message format

Method — what action to perform

| Method        | Action                          |
|---------------|---------------------------------|
| <b>GET</b>    | Retrieve a resource             |
| <b>HEAD</b>   | Retrieve headers only (no body) |
| <b>POST</b>   | Submit data to a resource       |
| <b>PUT</b>    | Store / replace a resource      |
| <b>DELETE</b> | Remove a resource               |

Request line

**method** URL version cr lf

Header lines

field name: value cr lf  
field name: value cr lf

Blank line

cr lf

Entity body

*entity body*

# HTTP request message format

**URL:** path to the requested resource — e.g.

`/someDept/pic.gif`

**HTTP Version:** identifies the protocol in use — HTTP/1.0 or HTTP/1.1

**Request line**

method

URL

version

cr lf

**Header lines**

field name:

value

cr lf

field name:

value

cr lf

**Blank line**

cr lf

**Entity body**

*entity body*

# HTTP request message format

## Header lines

- Zero or more `field-name: value` pairs, one per line
- Common examples:

```
Host: www.example.com
```

Mandatory in HTTP/1.1 — identifies the target server

```
Content-Type: text/html; charset=UTF-8
```

Format of the message body

```
Connection: close
```

Close the TCP connection after this response

```
Accept-Language: da, en-gb;q=0.8,  
en;q=0.7
```

Preferred languages, weighted by q (default 1): Danish → British English → any English

## Request line

method URL version cr lf

## Header lines

field name: value cr lf

field name: value cr lf

## Blank line

cr lf

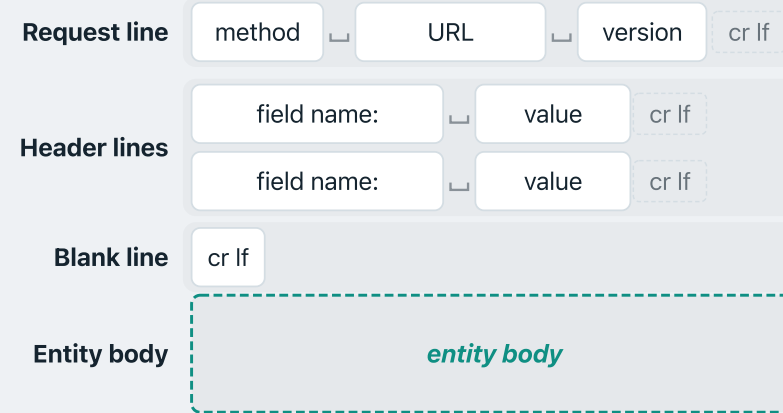
## Entity body

*entity body*

HTTP

# HTTP request message format

**Entity body:** empty for GET and HEAD; carries the payload for POST and PUT (e.g. form data or JSON)



HTTP

## HTTP request: an example

```
GET /somedir/page.html HTTP/1.1
```

```
Host: www.someschool.edu
```

```
User-agent: Mozilla/4.0
```

```
Connection: close
```

```
Accept-language: fr
```

-----  
*(empty entity body)*

**Method:** GET

**URL:** /somedir/page.html

**HTTP version:** HTTP/1.1

**Header field:** Host: www.someschool.edu

**Header field:** User-agent: Mozilla/4.0

**Header field:** Connection: close

**Header field:** Accept-language: fr

**Body:** (empty)

# HTTP response message format

## Version

The HTTP version the server is using — e.g. HTTP/1.1, just as in the request line.

Status line

version status code phrase cr lf

Header lines

field name: value cr lf

field name: value cr lf

Blank line

cr lf

Entity body

*entity body*

# HTTP response message format

## Status code & phrase

- **Status code:** 3-digit number indicating the result of the request
- **Phrase:** human-readable text accompanying the code — e.g. OK, Not Found

## Status line

version — status code — phrase — cr lf

## Header lines

field name: — value — cr lf

field name: — value — cr lf

## Blank line

cr lf

## Entity body

*entity body*

# HTTP response message format

## Common examples

| Code | Phrase                     |
|------|----------------------------|
| 200  | OK                         |
| 301  | Moved Permanently          |
| 400  | Bad Request                |
| 404  | Not Found                  |
| 505  | HTTP Version Not Supported |

### Status line

version — status code — phrase — cr lf

### Header lines

field name: — value — cr lf  
field name: — value — cr lf

### Blank line

cr lf

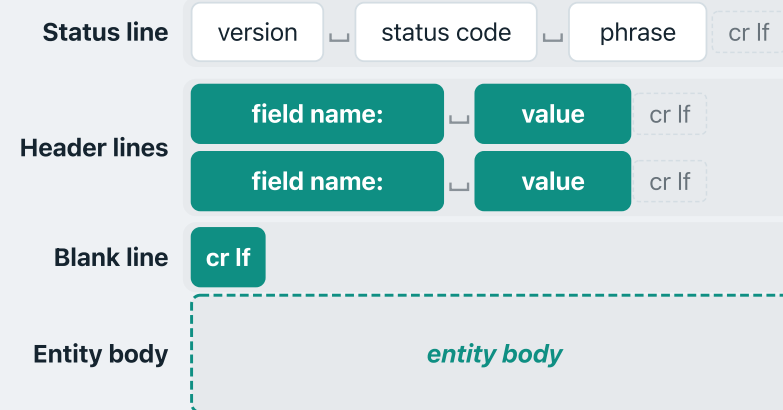
### Entity body

*entity body*

# HTTP response message format

## Header lines, blank line, entity body

Same structure as in the request: `field-name: value`  
header lines, a blank line (a lone CRLF), then the body —  
here, the **requested object** itself.



# HTTP response: an example

```
HTTP/1.1 200 OK
Connection: close
Date: Thu, 06 Aug 1998 12:00:15 GMT
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 1998 ...
Content-Length: 6821
Content-Type: text/html
```

```
<!DOCTYPE html>
<html><head><title>Example</title></head>
<body><h1>Hello!</h1>
<script>console.log("hi");</script>
</body></html>
```

**Version:** HTTP/1.1

**Status code:** 200 — success

**Phrase:** OK

**Header:** Connection: close — close after this response

**Header:** Date — when the response was sent

**Header:** Server — the server software

**Header:** Last-Modified — when the object last changed

**Header:** Content-Length — body size in bytes

**Header:** Content-Type — the body is HTML

**Body:** the requested HTML file (with embedded JS)

# Cookies

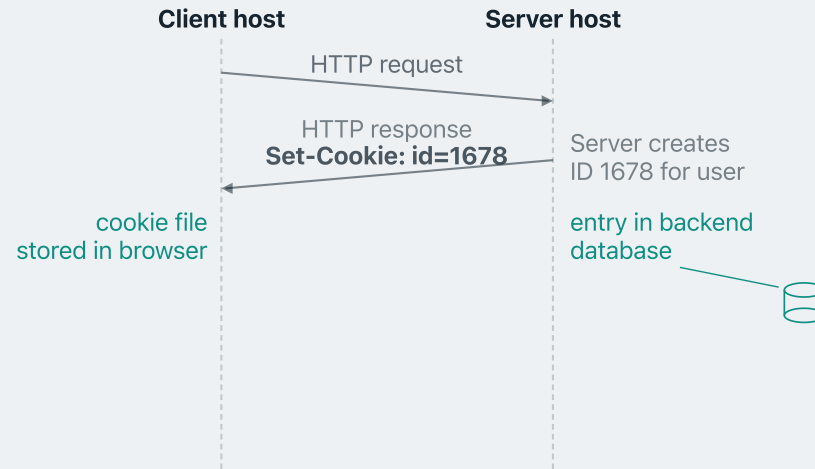
HTTP is **stateless** — every request is independent; the server retains no memory of past requests.

So the server does not remember your earlier conversations with it. Yet many applications *need* that memory: when you refresh an online shop's page, the items in your cart should not just disappear. **How does the server know what is in your cart?**

The answer: **cookies** — here's how it works.

1. A **Set-Cookie** header line in the **HTTP response** — the server sends the client a unique ID on first contact
2. A **Cookie** header line in subsequent **HTTP requests** — the client echoes the ID back on every future request
3. A **cookie file** stored on the client, managed by the browser — persists across sessions
4. A **back-end database** on the server, keyed by the cookie ID — stores per-user state

# Cookies: an example [RFC 6265]

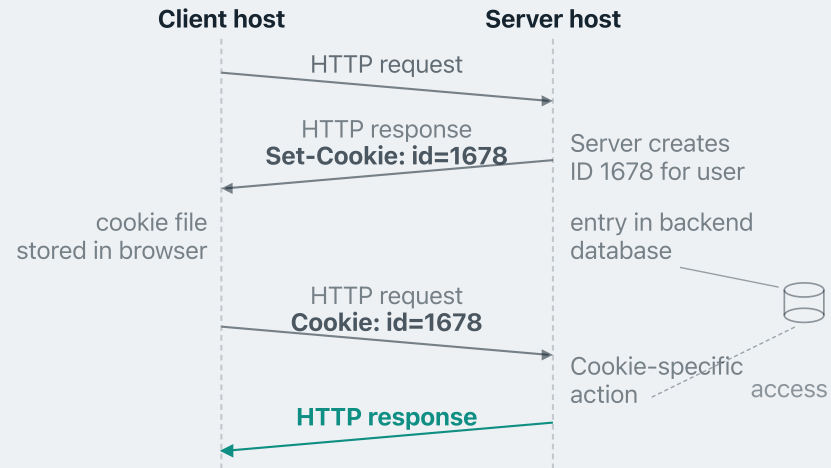


The client sends an HTTP request to the host

**Set-Cookie: id=1678** — Server assigns a unique ID and sends it to the client

The cookie ID is stored as an entry in the back-end database, and in the client's browser

# Cookies: an example [RFC 6265]



**Cookie: id=1678** — Client attaches the ID on every future request to the same server

Server uses the ID to look up the user's state in its database

Server can then generate a **tailored response** — e.g. a personalised page, a pre-filled shopping cart — for the client's request

HTTP

## One problem with the Web

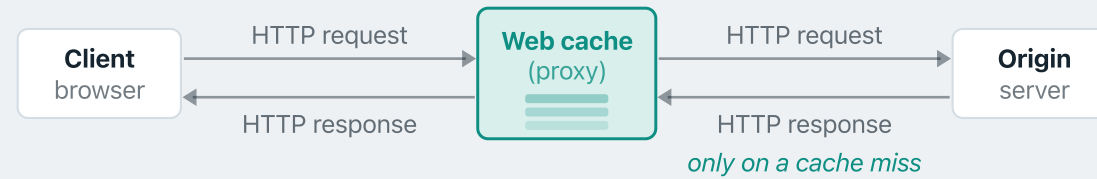
Picture a university: thousands of users fetch the same popular pages and images, and every request travels across the Internet to the **origin server** that hosts the object.

Every one of those requests lands on the same origin server, through the same congested links — enough of them can **overwhelm** it.

**How can we avoid this?**

# Web caching

A **web cache** (also called a **proxy server**) is a network entity that satisfies HTTP requests on behalf of an origin server — storing copies of recently requested objects locally.



## Acts as a server

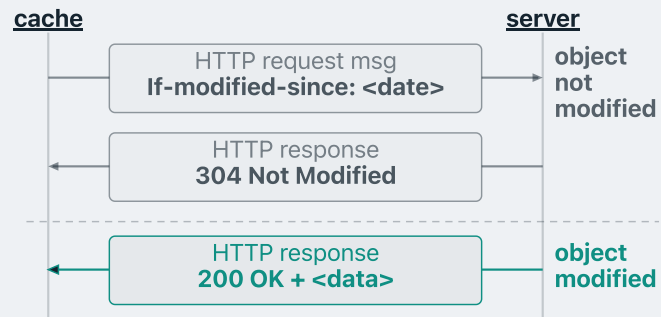
Receives HTTP requests from clients and, if it holds a copy of the object, returns it in the response — just like an origin server would

## Acts as a client

When the object is not cached, it issues its own HTTP request to the origin server, stores the copy, then answers the client

Requests answered by the cache get a **quicker response**, and caching **reduces how often** requests have to go to the origin server.

# Conditional GET



A **conditional GET** is an ordinary GET with an `If-modified-since` header line. A proxy uses it to check whether its cached copy is still up to date, e.g.:

```
GET /index.html HTTP/1.1
```

```
Host: www.example.com
```

```
If-modified-since: Mon, 10 Aug 2026 09:00:00  
GMT
```

If the object is **unchanged**, server replies:

```
304 Not Modified
```

Object **changed**: server replies 200 OK with the updated object in the body

MODULE 6

# Email

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Electronic Mail

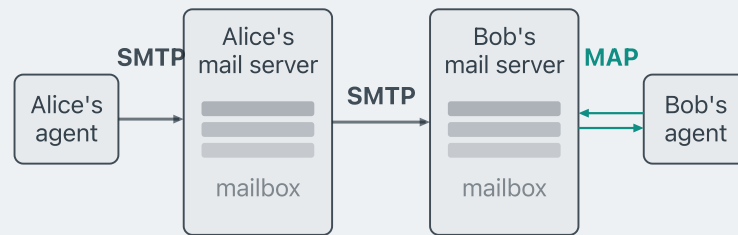
1 · Network applications

2 · The Web and HTTP

**3 · Email**

4 · DNS

# An overview of the email system



**User agent (UA)** — the app used to read, compose and send email (e.g. Outlook, Gmail, Apple Mail)

**Mail server** — a dedicated host that handles email on behalf of its users. It stores a **mailbox** per recipient for incoming mails and a **send queue** of outgoing mails

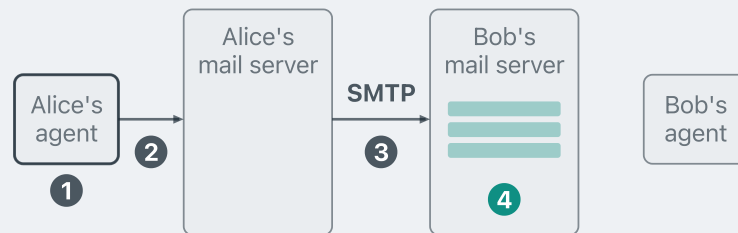
**SMTP** — Simple Mail Transfer Protocol; the sender's UA uses it to send a message to its own mail server

The sender's mail server then uses **SMTP** to transfer the message to the recipient's mail server across the internet

**Mail access protocol (MAP)** — used by UAs to retrieve messages from their mail server

## SMTP in action

SMTP has a **client side** and a **server side**, and always runs over **TCP**.



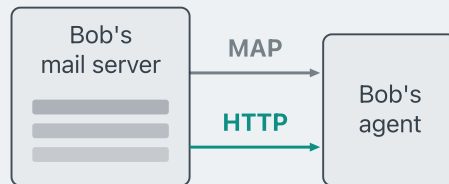
① Alice uses her UA to **compose** the message

② Alice's UA (**SMTP client**) **sends** the message to her mail server (**SMTP server**)

③ The **SMTP client** on Alice's server opens a TCP connection to Bob's mail server and **transfers** Alice's message over it

④ The **SMTP server** on Bob's server receives the message and places it in Bob's **mailbox**

## Retrieving email from the server

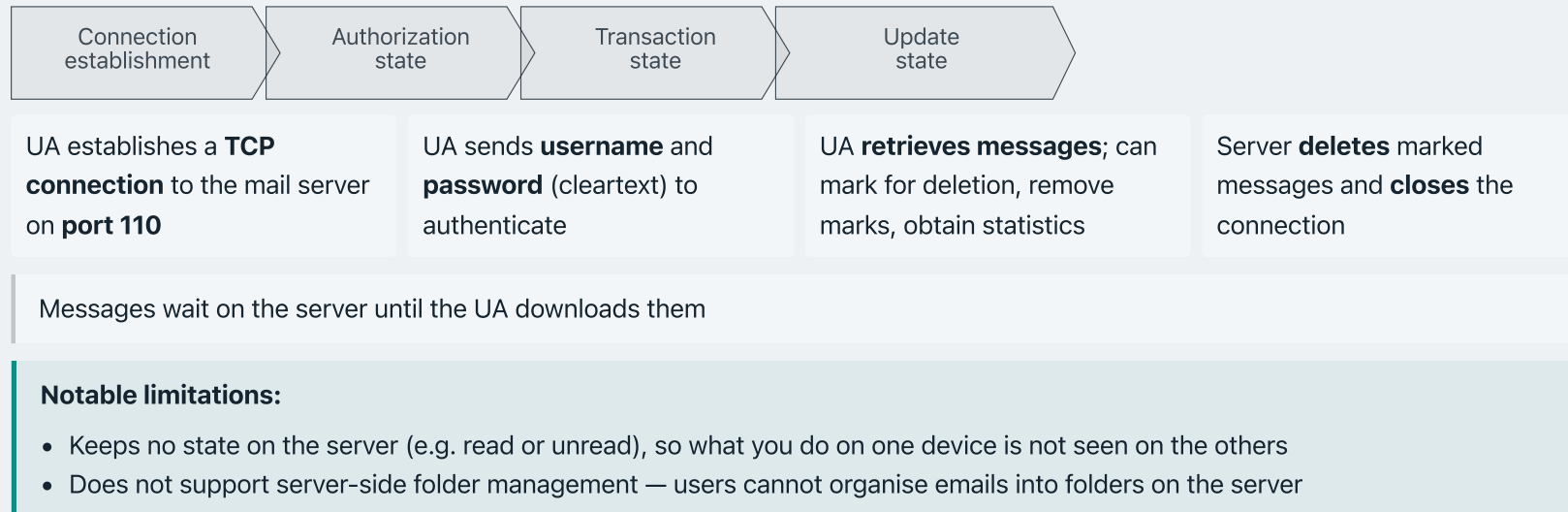


The recipient's UA retrieves email from its mail server either with a **mail access protocol (MAP)**, e.g.

- **POP3** — Post Office Protocol version 3 [RFC 1939]
- **IMAP** — Internet Message Access Protocol [RFC 9051]

or through **web-based email**, where the UA is a web browser that uses **HTTP** to reach the mail server

## How POP3 works



IMAP

# Internet Message Access Protocol

IMAP addresses POP3's limitations:

IMAP **keeps per-message state on the server**, e.g. read or unread, flagged, so every device sees the same status

It supports **folders on the server**: create folders and move messages into them, visible from any device

# Accessing email via a browser

## Web-based email is accessed through a web browser

The mailbox stays on the server; reading an email is like browsing the Web — the browser fetches each message as a web page

## Which application-layer protocol does the browser use to talk to the mail server?

**HTTP**, the same protocol behind every other web page

## Web-based email services

Examples: **Gmail**, **Outlook** (web)

## Comparing POP3, IMAP and web-based email

|                         | POP3                                                                  | IMAP                                                | Web-based email                                 |
|-------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------|
| <b>Accessed with</b>    | A mail app                                                            | A mail app                                          | A web browser, over <b>HTTP</b>                 |
| <b>Messages</b>         | Downloaded to the device; deleted from the server, or optionally kept | Kept on the server                                  | Kept on the server                              |
| <b>Folders</b>          | None on the server; a single mailbox                                  | Created and kept on the server                      | Created and kept on the server                  |
| <b>Multiple devices</b> | Not synced: read status and folders exist only on each device         | Read status and folders stay in sync across devices | Any device with a browser sees the same mailbox |

# Structure of an email message

An email message consists of **header fields**, followed (optionally) by a **body**

**Header fields** describe the message, e.g. **From:** and **Subject:**

The **body** carries the message content itself

The whole message is **ASCII text**, which encodes English letters, digits and punctuation

**Problem:** ASCII alone cannot carry images, attachments, or non-ASCII characters (e.g. Chinese, Arabic)

Date: Mon, 10 Aug 2026 09:00 GMT

From: alice@crepes.fr

To: bob@hamburger.edu

Subject: Picture of yummy crepe.

Cc: charlie@example.com

**Header fields**

---

Hello Bob, here is the picture  
you asked for. Enjoy!

**Body**

## MIME

# Multipurpose Internet Mail Extensions

**MIME** extends the message format with **additional header fields**

These fields declare the content's **type** and how it is **encoded**

so the receiver knows how to **decode the body**

e.g. **base64** encoding converts binary data into printable ASCII text, which can be used for image attachments

From: alice@crepes.fr

To: bob@hamburger.edu

Subject: Picture of yummy crepe.

**MIME-Version: 1.0**

**Content-Transfer-Encoding: base64**

**Content-Type: image/jpeg**

(base64 encoded data...  
...base64 encoded data)



decoded image

MODULE 6

# DNS

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Domain Name System

1 · Network applications

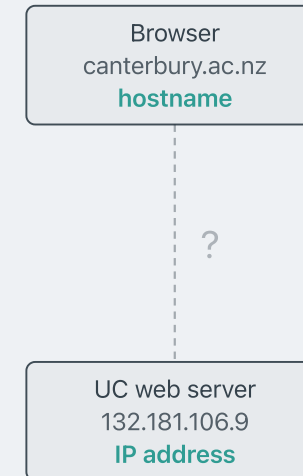
2 · The Web and HTTP

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4 · **DNS**

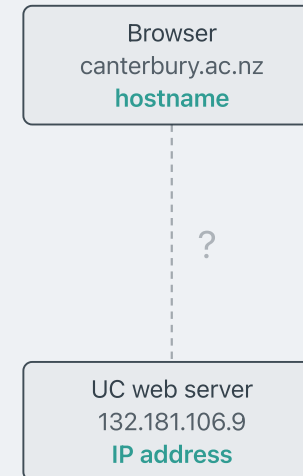
# Hostnames and IP addresses

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers, e.g. `132.181.106.9`
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?



DNS

# Domain Name System — overview



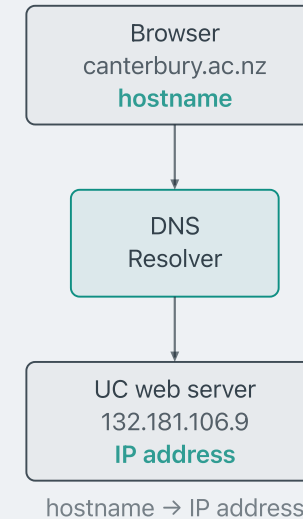
# Domain Name System — overview

**DNS** is that translation mechanism — it maps human-readable **hostnames** to numerical **IP addresses**

DNS stores this mapping in a **distributed database**, spread across a hierarchy of DNS servers [RFC 1034, 1035]

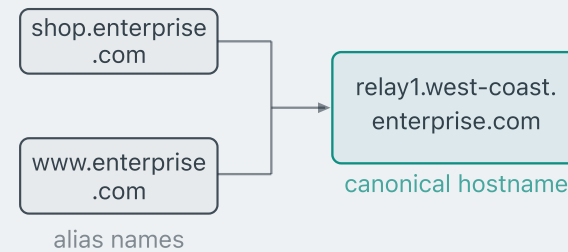
DNS is an **application-layer protocol** running mainly over **UDP on port 53** (TCP port 53 for large responses and zone transfers)

Why UDP? A query and its reply each fit in one small datagram, so there is no connection setup delay, providing a quick response.



## Additional services provided by DNS

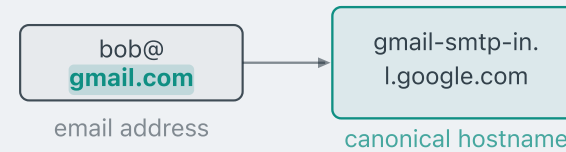
**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember. With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.



## Additional services provided by DNS

**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember. With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.

**Mail server aliasing** — the same idea for email. e.g. the mail server that handles Gmail has the cumbersome canonical hostname `gmail-smtp-in.l.google.com`. Yet Bob's email address can simply be `bob@gmail.com`: DNS uses the part after the @, `gmail.com`, to find that mail server.



# Additional services by DNS: load distribution

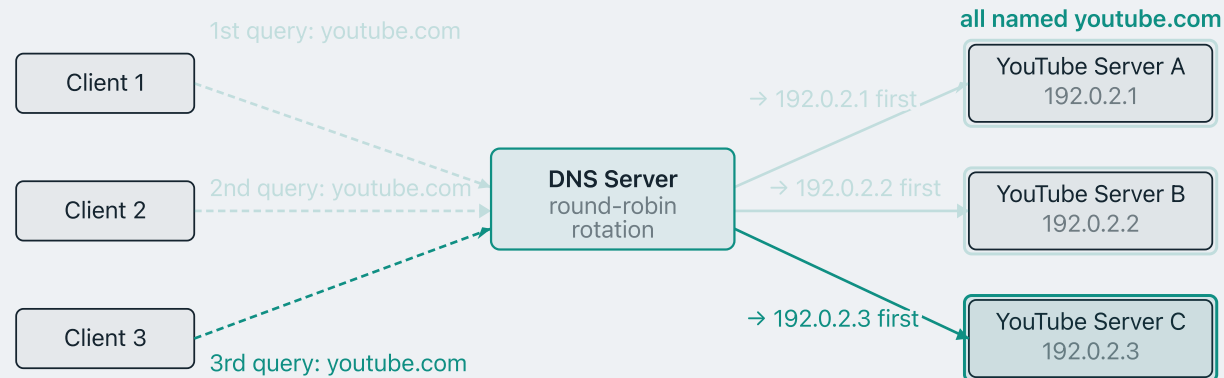
Many popular web services — e.g. YouTube, Google, Netflix — are hit by **millions of users** simultaneously. A **single server** cannot handle that volume.

The service is therefore distributed across **many server instances**, each with a **distinct IP address**, all sharing the **same hostname** (e.g. `youtube.com`).

**How are users assigned to these servers?**

**DNS load balancing** — DNS stores all the IPs and **rotates their order** on each response, so consecutive clients get a different IP listed first.

e.g. three clients query `youtube.com` in turn:



# DNS is distributed, not centralised

Recall that DNS stores the hostname-to-IP mapping in a **distributed database**, spread across many DNS servers.

**Why not a centralised design — a single server hosting the whole database, so that every DNS query goes to it?**

## Single point of failure

If the one server crashes, DNS lookup fails for the entire Internet

## Traffic volume

Billions of DNS queries per day — no single machine can handle that load

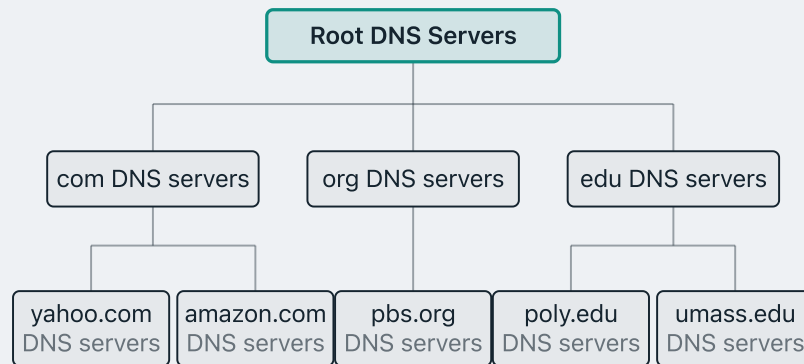
## Distant centralised database

All queries must travel to one location — users far away face high latency on every lookup

## Maintenance

Billions of hostname records change constantly — impossible to keep one database current

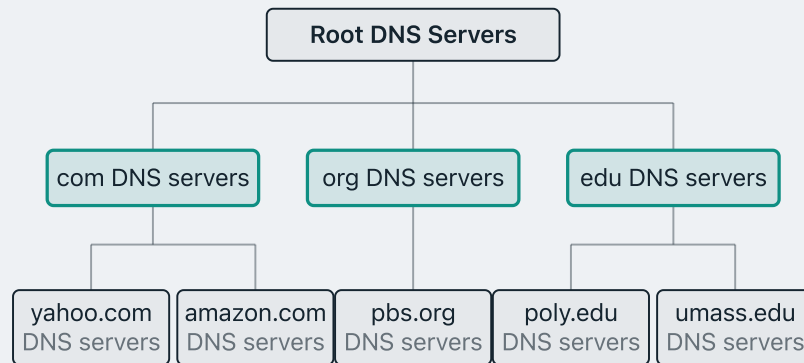
# A hierarchical distributed database



## Root DNS Servers

- 13 root server identities (a–m.root-servers.net)
- Each identity's IP address is shared by many machines worldwide; a query goes to the nearest one
- Together, **over 1000 physical instances**

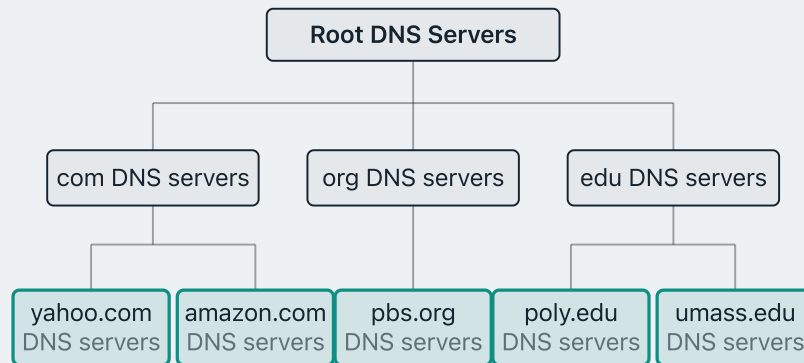
# A hierarchical distributed database



## Top-Level Domain (TLD) DNS Servers

- Handle generic domains: **.com**, **.org**, **.edu**
- ...and country-code domains: **.nz**, **.uk**
- Each is run by a dedicated registry, e.g. **Verisign** (.com), **InternetNZ** (.nz)

# A hierarchical distributed database



## Authoritative DNS Servers

- Each organisation's own DNS server
- The **definitive source** for its hostname-to-IP mappings
- e.g. Canterbury's authoritative server holds the IP for `www.canterbury.ac.nz`

# Local DNS server and DNS caching

Every ISP or institution operates a **local DNS server** (also called a *default name server*). It is **not** part of the root / TLD / authoritative hierarchy — it sits **separately** from that.

When you join the network, your device is automatically configured to use one — e.g. the University of Canterbury runs its own local DNS server for all campus users.

**DNS Hierarchy**  
Root · TLD · Authoritative  
(separate from local DNS)

**Local DNS server**  
e.g. dns.canterbury.ac.nz  
(cache)

Your computer

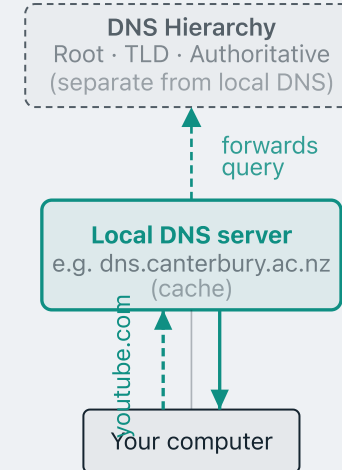
# Local DNS server and DNS caching

When your machine makes a DNS lookup, the query goes **first to the local DNS server**.

**If the answer is cached:** the local DNS server replies **immediately** — no hierarchy lookup needed.

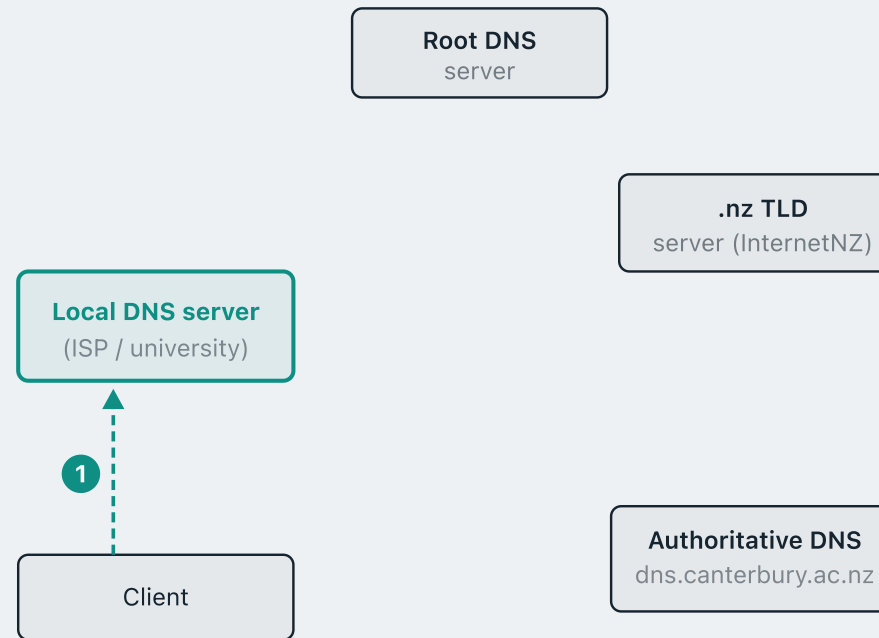
**If not cached:** the local server **forwards the query into the hierarchy**, relays the reply back to you, and **caches the hostname-to-IP mapping** in local memory for future queries.

Cache entries have a **TTL (time-to-live)** and expire after some time (typically hours). TLD server addresses are routinely cached in local DNS servers — so **root servers are bypassed for most everyday queries**.



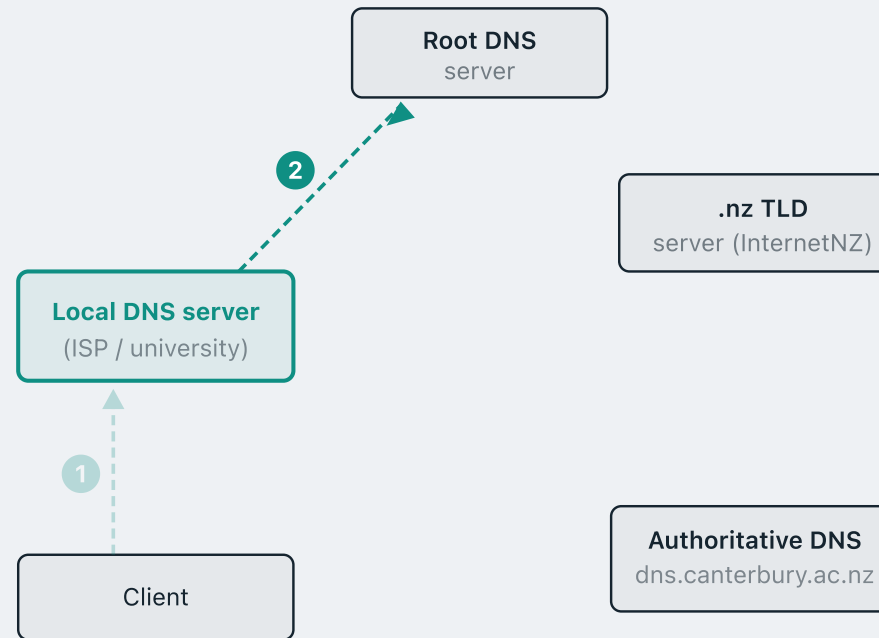
# Resolving a hostname: the full journey

**Step 1.** A client wants to resolve **csse.canterbury.ac.nz**. It sends a DNS query to its **local DNS server**.



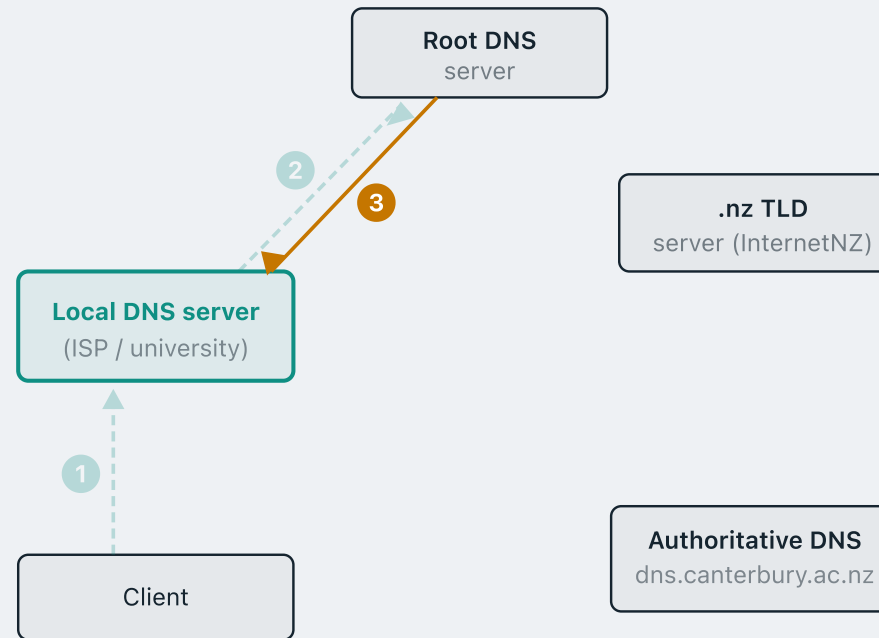
# Resolving a hostname: the full journey

**Step 2.** The local DNS doesn't have the answer cached. It contacts a **root DNS server**.



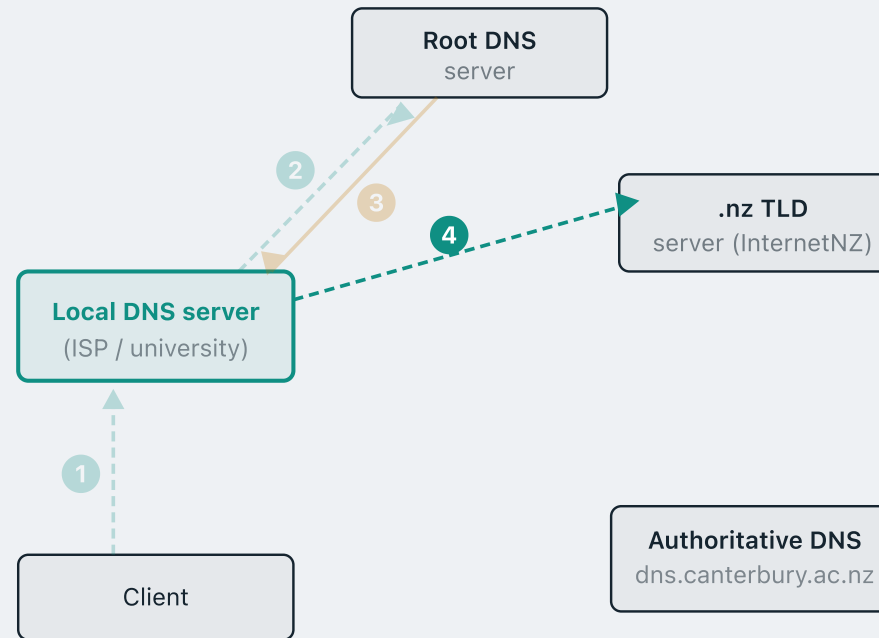
# Resolving a hostname: the full journey

**Step 3.** The root server returns a **referral**: "I don't know — try a **.nz TLD** server" and gives its address.



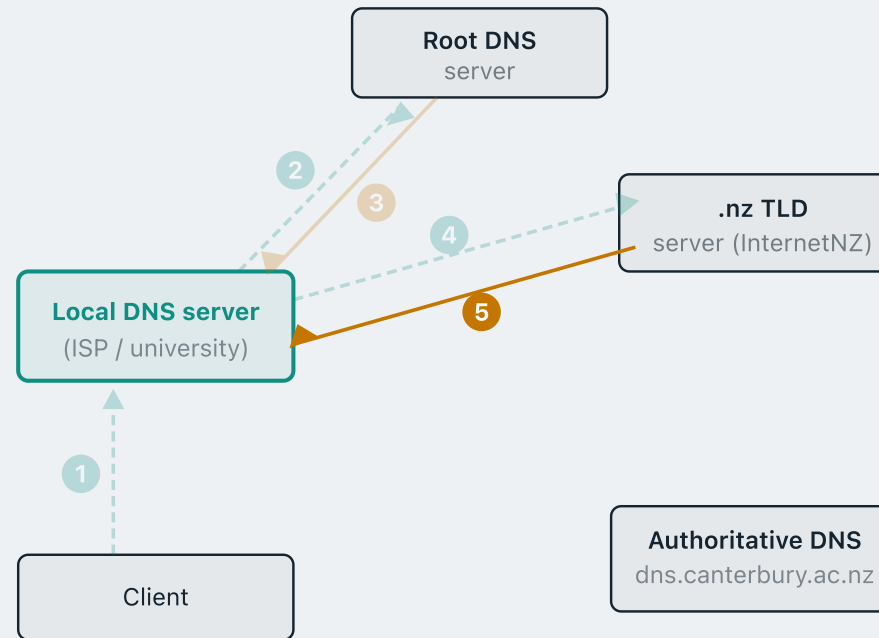
# Resolving a hostname: the full journey

**Step 4.** The local DNS follows the referral and contacts the **.nz TLD server**.



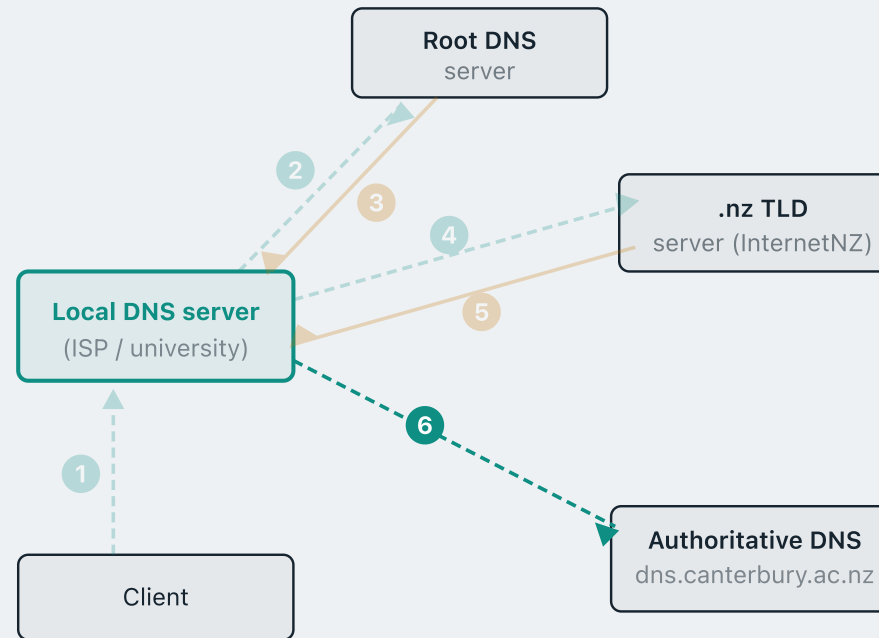
# Resolving a hostname: the full journey

**Step 5.** The .nz TLD returns another **referral**:  
"Try **dns.canterbury.ac.nz** — the  
authoritative server for canterbury.ac.nz."



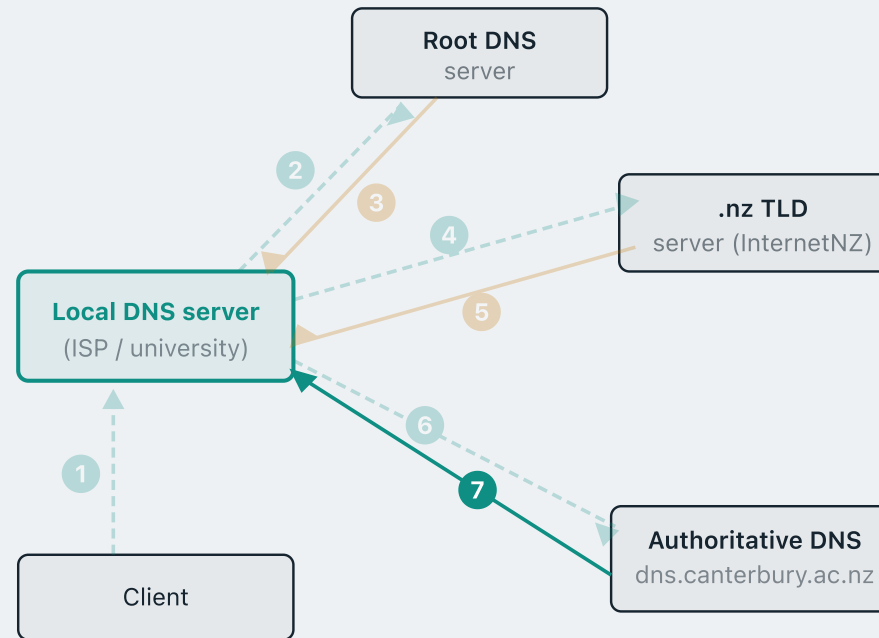
# Resolving a hostname: the full journey

**Step 6.** The local DNS contacts **dns.canterbury.ac.nz** — the authoritative name server for the canterbury.ac.nz zone.



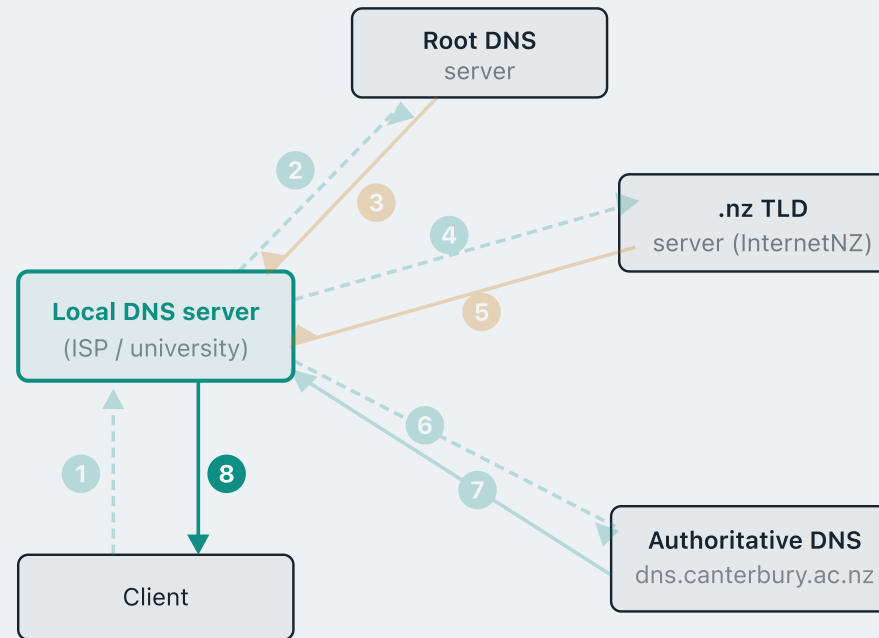
# Resolving a hostname: the full journey

**Step 7.** The authoritative server has the answer! It replies: "csse.canterbury.ac.nz = 132.181.2.21"



# Resolving a hostname: the full journey

**Step 8.** The local DNS returns **132.181.2.21** to the client — and **caches** the mapping for future queries.



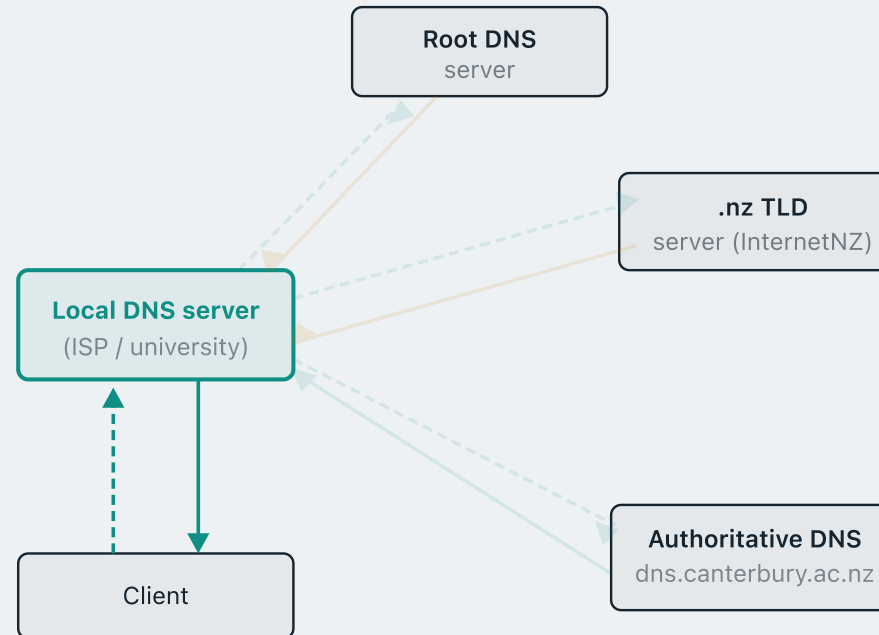
# Recursive vs iterative queries

## Iterative query

The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

## Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.



# Recursive vs iterative queries

## Iterative query

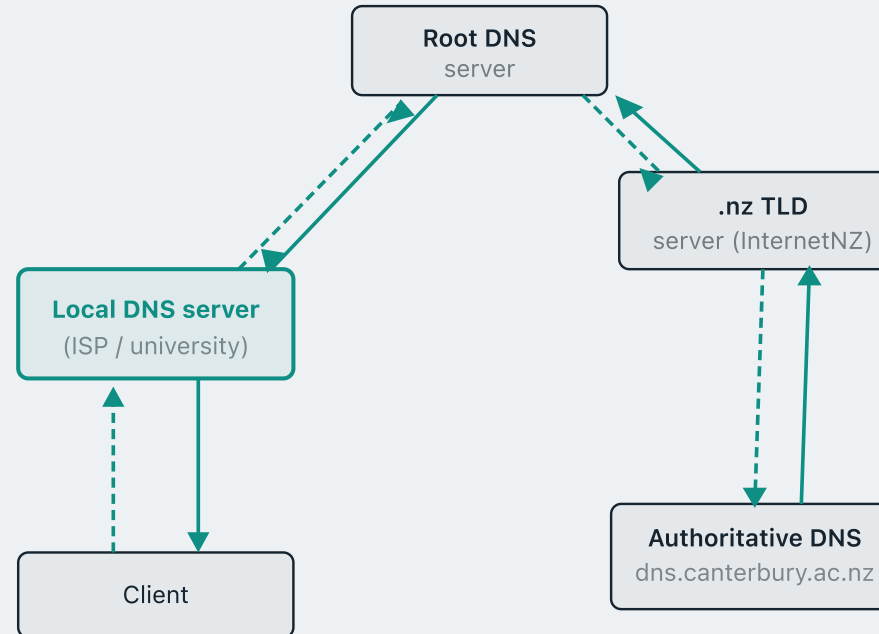
The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

## Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.

## Can we go fully recursive?

In principle, yes — root asks TLD, which asks auth. In practice, root and TLD servers refuse recursion (too much load), so only client → local DNS is recursive.



# DNS Resource Records

DNS distributes data as **Resource Records (RRs)**. Each record has four fields:

(name, value, type, TTL – Time-To-Live)

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type A

**name** = hostname · **value** = IPv4 address

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

## Type NS

**name** = domain · **value** = hostname of the authoritative DNS server for that domain

*"I don't know — but here's the name of the server that does."*

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type CNAME

**name** = alias hostname · **value** = canonical (real) hostname it maps to

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

## Type MX

**name** = domain · **value** = hostname of the mail server for that domain

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS protocol message format

DNS uses the **same message format** for both **queries** and **replies** — a fixed 12-byte header followed by four variable-length sections.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Identification**

16-bit number assigned by the client. The reply copies this same value, so the client can match each reply to its original query.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Flags**

- Query (0) or reply (1)
- Recursion desired — client requests full recursive resolution
- Recursion available — server indicates it supports recursion

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Count fields**

Four 16-bit counters — one per section below — telling how many records appear in each variable-length section.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Questions**

The name and record type being queried. e.g. this  
Type A question asks for the IP address of the domain  
facebook.com.

( facebook.com, A )

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

Answers

Resource Records (RRs) that directly answer the query.

```
(facebook.com, 157.240.22.35, A, TTL=300)
```

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Authority**

NS records pointing to authoritative servers —  
returned in referral responses.

(facebook.com, a.ns.facebook.com, NS)

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Additional**

'Glue' Resource Records (RRs) accompanying the authority section — so you can reach the name server without an extra lookup.

(a.ns.facebook.com, 129.134.30.12, A)

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# Query & response: a live example

## QUERY →

Transaction ID 0x6492  
Flags 0x0100 (query)  
Questions 1 Answer RRs 0  
Authority RRs 0 Additional RRs 0

### QUESTIONS

www.facebook.com A

## ← RESPONSE

Transaction ID 0x6492  
Flags 0x8180 (response, no error)  
Questions 1 Answer RRs 2  
Authority RRs 4 Additional RRs 8

### QUESTIONS

www.facebook.com A

### ANSWERS

www.facebook.com CNAME  
star-mini.c10r.facebook.com  
star-mini.c10r.facebook.com A  
31.13.78.35

### AUTHORITY

c10r.facebook.com NS  
d.ns.c10r.facebook.com (+3)

### ADDITIONAL

d.ns.c10r.facebook.com A  
185.89.219.11 (+7)

# DNS — key takeaways

- **Main purpose** — DNS maps human-readable hostnames to IP addresses.
- **Hierarchy** — DNS is distributed: no single server holds all records.  
Responsibility is delegated across Root, TLD, and Authoritative servers.
- **Traversal pattern** — the client queries the local DNS resolver, which then follows referrals Root → TLD → Authoritative.  
Results are cached locally by TTL to avoid repeated lookups.

## References

James F. Kurose & Keith W. Ross, *Computer Networking: A Top-Down Approach*, Addison Wesley, 3<sup>rd</sup> edition, 2005.

Chapter 2 Application Layer (2.1, 2.2, 2.4, 2.5)

William Stallings, *Data and Computer Communications*, Pearson, 10<sup>th</sup> edition, 2013.

Chapter 24 Electronic Mail, DNS, and HTTP (24.1–24.3)

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5<sup>th</sup> edition, 2010.

Chapter 7 The Application Layer (7.1–7.3)

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